

ICT702 Data Visualisation

Explaining Visually to Influence with Data Module 7



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



Part 2

Empathizing With Data

- **Empathy** refers to the ability to understand and share the feelings of others.
- By empathizing with the data, we can create data visualisations that impact decisions effectively.
- The two common challenges for creating empathy with data are:
 - Audiences can lose the ability to associate meaning when considering large numerical values.
 - It can be challenging to consider individual cases when looking at aggregate statistics.

Strategies for interpreting Large Numbers

Convert a large number into something with which the audience may be more familiar.

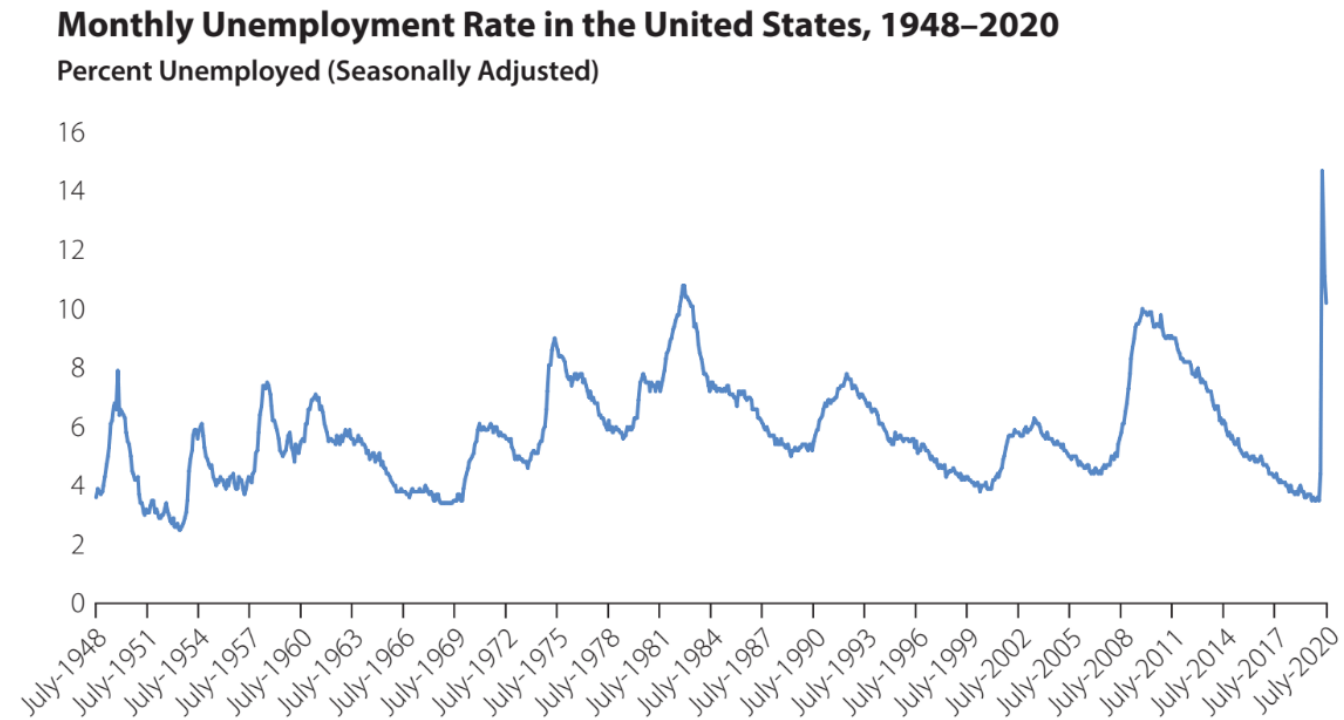
- Show a Powerball Lottery payout of \$357 million, as \$171,000 per week for the next 40 years.
- To convey exponential growth, replace use of large numbers with sentences like “The last two years saw the creation of 90% of all the data in existence.”

Avoid the use of scientific notation for large numbers in data visualisation.

- Use words such as “millions” and “billions” rather than 10^6 and 10^9 .
- It is difficult to convey the magnitude of change in charts without proper context.

Empathizing With Data: a Simple Line Chart

Simple line chart showing monthly unemployment rate in US, 1948 to 2020



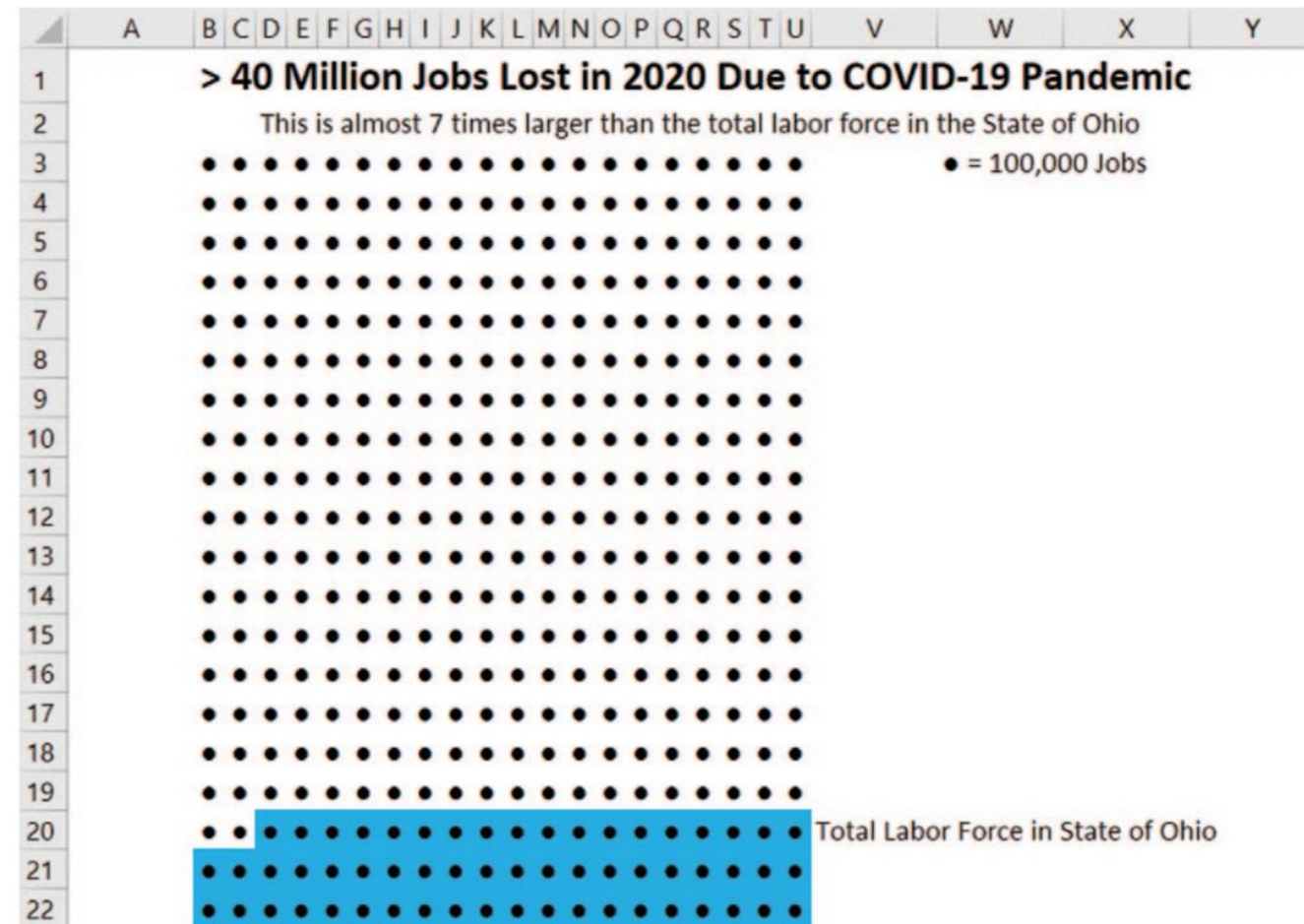
(based on data from U.S. Bureau of Labor Statistics, <https://data.bls.gov/>)

- The unemployment rate has increased substantially
- Difficult for audience to interpret the magnitude of this change without additional context

Empathizing With Data: a Dot Matrix Chart

A dot matrix chart illustrating number of jobs lost in US during the 2020 COVID-19 pandemic relative to total labor force in Ohio.

- The dots are laid out in a matrix form.
- The size of the matrix is relative to the size of the total number to be conveyed.

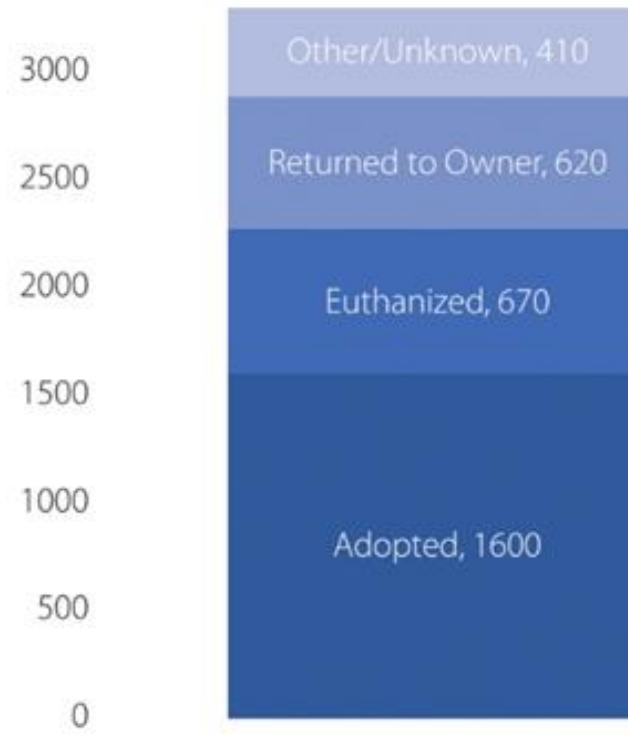


Empathizing With Data: Add a Picture

A stacked column chart that includes a picture of a specific shelter dog creates more empathy with the data

To add a picture to an existing chart in Excel, click the **Insert** tab on the **Ribbon**, click the **Pictures** button in the **Illustrations** group, and then choose from where you want to take the picture.

Fates of the Dogs Who Enter Animal Shelters Each Year
Number of Dogs (in 1000s)



This is shelter dog Max.

Max entered the animal shelter in Portland in April and he is still available for adoption today.

Empathizing With Data: a Clustering Algorithm

Summary results from clustering algorithm used for customers of Third State Bank

Customer Characteristic	Cluster #7 Results
Average Age	24
Education Level	67% college or above
Marital Status	72% unmarried
Average Number of Children	0.3
Home Location	67% urban, 21% suburban, 12% rural
Closely Follows Sports?	34% Yes, 66% No
Closely Follows Politics?	53% Yes, 47% No

In market segmentation analysis, clustering algorithms divide a company's customers into different groups (clusters) that have similar characteristics.

End of Part 2

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